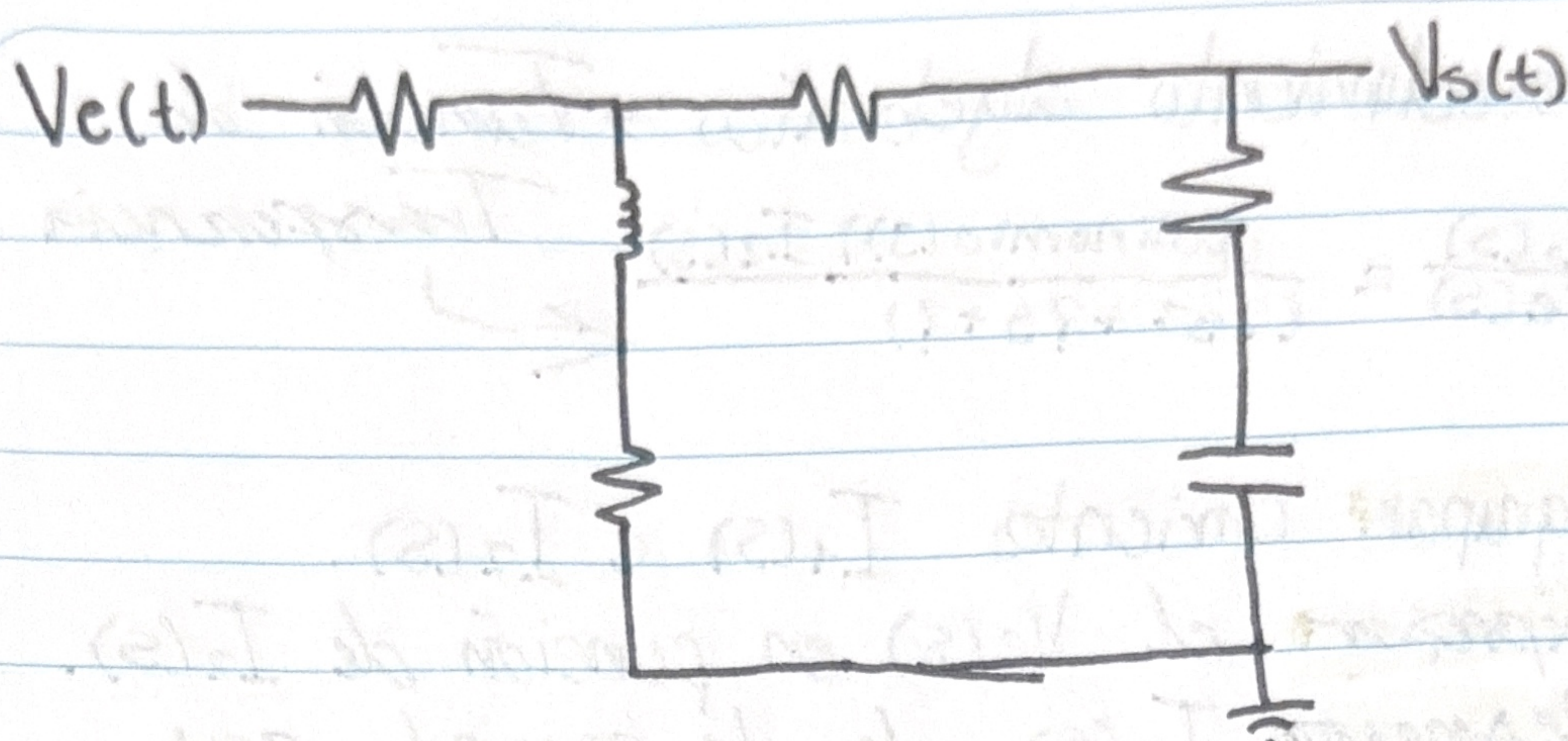


5.1 Práctica: Diseño de controlador para un sistema de 2^{do} orden



Ecuaiones princ.

$$V_c(t) = R i_1(t) + L \frac{d(i_1(t) - i_2(t))}{dt} + R [i_1(t) - i_2(t)]$$

$$L \frac{d(i_1(t) - i_2(t))}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_c(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico $\rightarrow$ Función de Transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{(\text{polinomio}(s)) I_2(s)}{(?s^2 + ?s + ?)}$$



Agrupar corrientes $I_1(s)$ e $I_2(s)$.

Expresar el $V_e(s)$ en función de $I_2(s)$.

Despejar $I_1(s)$ de la segunda ecu. principal.

$$V_s(s) = (R + \frac{1}{Cs}) I_2(s) = (\frac{CRs + 1}{Cs}) I_2(s)$$

$$Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s) = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs} + Ls I_2(s) + R I_2(s)$$

$$(Ls + R) I_1(s) = (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$I_1(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_e(s) = R I_1(s) + Ls I_1(s) - Ls I_2(s) + R I_1(s) - R I_2(s)$$

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$V_e(s) = (Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) - (Ls + R) I_2(s)$$

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$$V_e(s) = \left[\frac{(Ls+2R)(CLs^2+3CRs+1)}{CLs^2+CRs} - (Ls+R) \right] I_2(s)$$

$$V_e(s) = \left[\frac{(Ls+2R)(CLs^2+3CRs+1) - (Ls+R)(CLs^2+CRs)}{CLs^2+CRs} \right] I_2(s)$$

$$V_e(s) = \left[\frac{CL^2s^3 + 3CRLs^2 + Ls + 2RCRs^2 + 6CR^2s + 2R}{CL^2s^3 + LCRs^2 + RCLs^2 + CR^2s} - \frac{CL^2s^3 + LCRs^2 + RCLs^2 + CR^2s}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \left[\frac{5RCLs^2 - 2RCLs^2 + 5CR^2s + Ls + 2R}{CLs^2 + CRs} \right] I_2(s)$$

$$V_e(s) = \frac{3RCLs^2 + 5CR^2s + Ls + 2R}{CLs^2 + CRs} I_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs+1}{Cs} I_2(s)}{\frac{3RCLs^2 + (5CR^2+L)s + 2R}{CLs^2 + CRs} I_2(s)}$$

$$= \frac{\frac{CRs+1}{Cs}}{\frac{3RCLs^2 + (5CR^2+L)s + 2R}{Cs(Ls+R)}}$$

Error en estado estacionario cuando se aplica un escalón unitario de entrada

$$e(s) = \lim_{s \rightarrow 0} s \frac{1}{s} \left(1 - \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR s^2 + (5CR^2 + L)s + 2R} \right) = \frac{R}{2R}$$

$$e(t) = 0.5 V$$

Estabilidad en lazo abierto

Se calculan los polos del sistema (raíces del denominador)

$$R = 10 k \Omega$$

$$L = 680 \mu F$$

$$C = 1000 \mu F$$

$$a = 9.588 \times 10^{-3}$$

$$b =$$

$$a = 2.04 \times 10^{-3}$$

$$c = 9.4 \times 10^{-3}$$

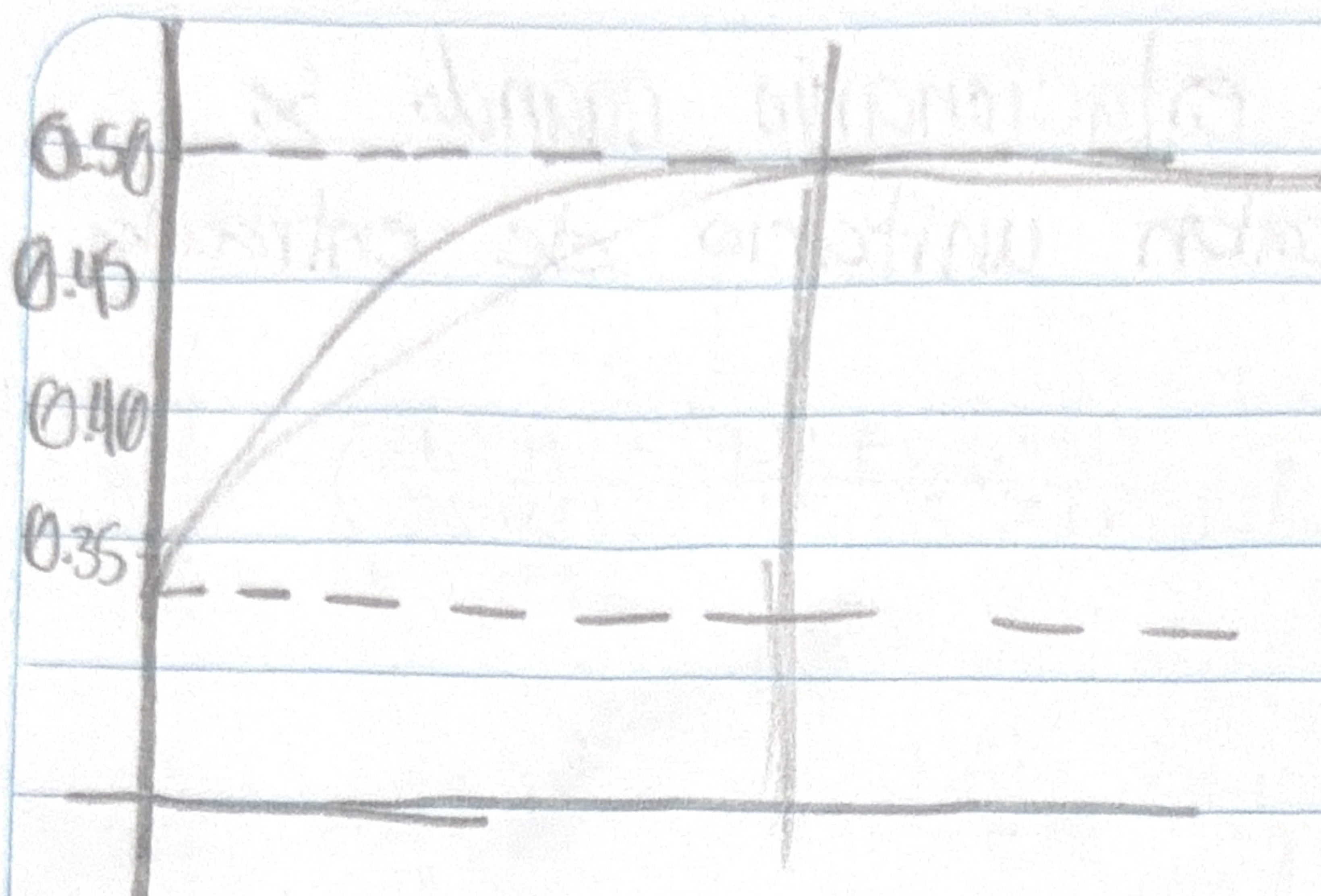
$$b = 9.00068$$

$$5003.4 \times 10^{-3}$$

$$c = 20 \times 10^3 \quad 200$$

$$\lambda_1 =$$

$$\lambda_2 =$$



$$\lambda_1 = -0.4$$

$$\lambda_2 = -2.45 \times 10^7$$

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Modelo de ecuaciones integro-diferenciales

Apartir de las ecuaciones principales del Sistema, se despejan las variables dependientes, es decir, las corrientes y voltajes de salida:

$$\begin{aligned} i_1(t) &= [V_e(t) + R i_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt}] \frac{1}{2R} \\ i_2(t) &= [L \frac{d[i_1(t) - i_2(t)]}{dt} + R i_1(t) - \frac{1}{C} \int i_2(t) dt] \frac{1}{3R} \\ V_s(t) &= R i_2(t) + \frac{1}{C} \int i_2(t) dt \end{aligned}$$

$$R = 100 \Omega$$

$$L = 680 \mu\text{Henrios} = 680 \times 10^{-6} \text{H}$$

$$C = 100 \mu\text{Faradios} = 100 \times 10^{-6} \text{H}$$

$$\begin{aligned} \frac{V_s(s)}{V_e(s)} &= \frac{CLRs^2 + (CR^2 + L)s + R}{3CLR^2 + (SCR^2 + L)s + 2R} = \frac{6.8 \times 10^{-6} + 1.00068}{204 \times 10^{-5} s^2 + 5.00068} \\ &+ 100 \\ &+ 200 \end{aligned}$$